Computational Modeling of Human Experience Formation through Practice: Evidence, Revision, Reuse, and Host-Agent Validation

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Abstract— Experience is not acquired merely by saving a piece of text. It emerges from goal-directed action, action outcomes, interpretation, counterexample testing, and subsequent reuse. Omitting any of these steps may turn an accidental success into an apparent rule. This paper models this human process as a computational cycle of practice, observation, interpretation, validation, experience, and renewed practice. Experience is represented by an object containing a claim, conditions, version, provenance, evidence, counterexamples, and lifecycle state. Candidate, stable, disputed, and invalidated knowledge are distinguished. Persistent version chains, provenance constraints, conflict records, explicit unknown outcomes, and host-context recovery allow applicability to be checked before the next action. Beta–Bernoulli updating describes changes in belief under limited evidence, information gain ranks informative verification actions, and Thompson sampling serves only as an optional exploration controller; the overall system is not characterized as Bayesian reinforcement learning. We first explain the experience-formation process, then present state transitions, update and recovery algorithms, and complexity, followed by mechanism experiments with 10 seeds, 12 rounds, and seven conditions, a 20-seed end-to-end gate experiment, and separate 10-seed Bayesian bandit controls. In the rule-reuse benchmark, B2 improves completion by 4.51 percentage points over an exact-index baseline and reduces source reads by 81.60%, while increasing local logical operations by 41.53% and providing no latency advantage. The end-to-end gate experiment places experience formation and task benefit in one ledger: with the candidate gate returned to the loop, completion remains 4.20 percentage points above exact indexing and source reads fall by 73.33%, at the cost of 12 additional reads per seed; the state gate pays off only on bad claims that have not yet completed a second independent practice (0 errors versus 209 errors). Real-host validation has a real model decide for itself, over the MCP protocol, between using experience and reading the source; seven pre-registered arms total 912 answers. Version admission and bad-experience detection hold. When correct content is mis-judged as disputed by a single failed observation, answers never become wrong, but the entire reuse benefit is zeroed and does not self-heal (the success count climbs from 3 to 6 while the state never leaves disputed); content-level revision with fresh evidence restores reuse; and with 300 same-family irrelevant memories also in the store, each recall still returns only the target entry. Under continuous drift, ablating the version and status metadata from recall drops accuracy from 12 of 12 to 3 of 12 for the same host and the same memories, with all nine wrong answers exactly equal to the obsolete-rule answer, while the non-drifting control group is 12 of 12 in both arms. Replaying the same ablation with two external API models as the host over 96 answers reproduces all four cells exactly, and the difference also holds in a clean subgroup whose content contains no version character at all. A final arm compares workflows rather than components, pitting a library-equipped arm (recall plus on-demand source reads) against an arm with no memory layer at all that can only read the source: with unlimited reads the two arms are equally accurate while the library arm needs 20 source reads instead of 72 and runs at 32% of the prompt tokens, and when each arm is capped at 6 or 12 source reads the library arm completes 18 and 24 of 24 tasks against 6 and 12, identically on two hosts from different families; the staleness gate makes zero wasted and zero missed reads across three drift levels, and when the budget is exhausted both arms return unknown instead of guessing; a seventh arm empties the library so that the host must file and revise its own entries, where both arms stay equally accurate over 72 questions, source reads fall from 72 to 36 (48 to 12 once the library exists), prompt tokens reach 59.2 percent of the opponent, and 24 filed entries plus 12 revisions match the manual revisions one for one with zero contamination and zero stale reuse. These results support experience reuse within bounded contractual tasks; they do not demonstrate self-evolution of model parameters or general improvements in language-model capability.

Index Terms— Practical experience; experience lifecycle; evidence constraints; version revision; host agent; Bayesian evidence updating.

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I. Introduction: Experience as Knowledge Tested through Practice

When addressing a problem, a person generally begins with a goal and acts using existing knowledge. Action produces an outcome, and comparison with the expected outcome yields feedback. Feedback does not automatically become experience: the outcome may be accidental, conditions may be undocumented, observations may be misinterpreted, or a formerly successful method may fail in a different environment version. The essential property of experience is the organization of action outcomes into knowledge that can be reused under explicit conditions, challenged, and revised. Agent-memory systems often emphasize storage and retrieval while leaving the validity, expiration, and correction of remembered knowledge less explicit [1-2].

This paper studies computational modeling of experience acquired through practice. The system, Butian, runs within a host AI application. The host supplies the model, sessions, and tools; Butian manages experience objects, evidence chains, version admission, state transitions, feedback persistence, and context recovery. Software memory is not equated with human consciousness, and textual records are not equated with parameter learning. Human practical reasoning is abstracted into an observable engineering process, whose usefulness is assessed through states, evidence, recurring errors, reuse benefits, and recovery quality.

A. Research Questions and Contributions

RQ1 asks whether the action–outcome–reflection–validation–reuse process can be represented as a state machine preserving evidence and conditions. RQ2 asks when candidate experience may become stable and when the system must return unknown or record a dispute. RQ3 asks whether the mechanism reduces repeated evidence acquisition and avoids obsolete-rule errors under version changes, source outages, and service restarts. RQ4 concerns the distinct roles of Bayesian updating and exploration strategies. The contributions are a layered

definition of practical experience, algorithms from evidence production to revision, multi-round mechanism experiments using the real ButianService, independent numerical controls, and explicit boundaries between host-AI evidence and mechanism experiments.

B. Related Work and Positioning

Experiential-learning research connects concrete experience, reflection, conceptualization, and subsequent experimentation. The systematic review of Kolb's model emphasizes context and reflection [3]. Accordingly, we distinguish having performed an action from having formed reusable experience: an individual occurrence is an event, an explanation is a candidate claim, and subsequent practice supplies further support or counterexamples. This relationship informs the research design; it does not imply a software threshold of two successes. That threshold is an auditable engineering choice requiring sensitivity analysis in specific task families.

Agent research provides complementary references. Reflexion converts task feedback into verbal reflection, and Voyager supports subsequent exploration through reusable skills [4-5]. MemoryBank studies long-term memory, while LongMemEval evaluates cross-session memory [1-2]. These studies do not independently establish the correctness of our thresholds or version rules. Here, provenance, nonduplicated evidence, renewed verification after version changes, and preserved disputes are placed within one practical process and examined separately. Retrieval-augmented generation provides access to external information [6], but retrieving relevant text still leaves an applicability decision to be made.

C. Organization

Section II defines the events within a practical episode. Section III explains candidate experience, evidence, counterexamples, versions, and reuse admission. Section IV presents algorithms and implementation. Sections V and VI describe experiments and results. Section VII discusses the roles of Bayesian components, engineering benefits, and limitations. Section VIII concludes. Mathematical tools support, rather than replace, the central account of experience formation.

II. A Process Model of Experience Formation through Practice

A. From Goals to Action Outcomes

At time t, let a person face goal G_t, environment E_t, and existing knowledge M_t. The person proposes action A_t, changes the environment through tools or physical action, and obtains observation O_t. The difference between the expected observation $\hat{O}_t$ and actual observation produces feedback $F_t = O_t - \hat{O}_t$. This is not merely a correct/incorrect label. An unexecuted action has no interpretable outcome; success cannot be transferred when the environment version is unknown; observations without provenance cannot be audited; and one occurrence does not establish statistical stability. The model therefore retains actions, environment snapshots, observation provenance, and outcome evidence rather than a summary alone.

$$(G_t, E_t, M_t, A_t) \rightarrow O_t \rightarrow F_t \tag{1}$$

B. From Feedback to Candidate Knowledge

Reflection asks what happened, why it happened, and under which conditions it may be reused. Candidate experience e contains claim h, applicability conditions c, environment version v, provenance s, evidence set E, and counterexample set X. A candidate is a testable interpretation, not an established fact. Outcomes without conditions create overly broad candidates; interpretations without provenance cannot be audited; deleting counterexamples obscures applicability boundaries.

$$e = (h, c, v, s, E, X, l, \theta) \tag{2}$$

Here l is lifecycle state, and θ summarizes evidence-supported belief. It is an estimate under the current evidence and working prior, not the probability of truth. A candidate may progress to warm only after further successful predictions in practice and in the absence of unresolved counterexamples.

C. Renewed Practice and Revision

A subsequent task first checks whether its conditions satisfy c and v. After adoption, the first-action outcome is frozen so that later answers cannot contaminate first-attempt evaluation. A successful prediction adds an observation with an independent key and evidence reference. Failure adds a counterexample and initiates dispute or revision. When sources, tools, or termination status cannot be confirmed, the result is unknown. Revision creates a new version with its parent, reason for change, and evidence rather than overwriting the earlier account.

D. Running Example: Formation of a Calibration Rule

Consider a sensor using a linear conversion in version v1. The first task specifies the entity, software version, and raw measurement. The host selects a rule from source material and produces an output; an instrument-calibration record supplies outcome evidence. Instead of recording a universal rule, the system creates the candidate that rule h is effective for entity a, version v1, and the stated measurement range. A second success with a different calibration sample and evidence digest satisfies the local threshold. Different files may nevertheless originate from the same erroneous source: deduplication removes recognized duplicates but cannot guarantee statistical independence.

After an update to v2, the old conditions no longer cover the task. The system should seek v2 evidence and retain unknown if that evidence is unavailable. Once sources recover, it creates a revision and tests it in further practice. Failure under genuinely matching v1 conditions attaches a counterexample to the original claim and changes its state to disputed; a more confident explanation cannot erase that failure. This example connects goals, actions, evidence, candidates, reuse, drift, and revision as one process.

E. Five Essential Stages

The process requires goal specification, action, observable outcomes, validation, and renewed practice. Without a target problem, experience has no task of application. It must constrain actual action, retain outcome evidence, check conditions and counterexamples, and be tested again in later tasks. Butian's state machine, evidence tables, version tables, recovery mechanism, and feedback interfaces implement observable counterparts of these stages. Omitting a stage leaves

a log, a retrieval result, or a candidate claim rather than a complete experience process.

III. Experience Lifecycle: From Candidates to Reusable Knowledge

A. States

candidate denotes experience from one or a few episodes that has not met stable-use conditions. warm denotes experience meeting the specified success and evidence thresholds and eligible for retrieval under matching conditions. hot denotes recently effective and frequently used experience; frequency is not additional proof of correctness. disputed records counterexamples, conflicts, or environmental mismatch and is not unconditionally reusable. invalidated records a claim known to no longer hold. unknown describes an unresolved observation or task outcome, not a memory-temperature state alongside candidate. The implementation also includes maintenance states such as cold, archived, and pending_validation. These names describe program records, not psychological states.

$$s \geq 2, f = 0: \text{candidate} \rightarrow \text{warm} \tag{3}$$

$$\text{failure: warm} \rightarrow \text{disputed} \tag{4}$$

B. Entry Requirements

External articles and host responses can supply candidate sources but cannot directly establish stable local experience. Entry requires a claim, applicable task, source event, environment or version information, and a traceable outcome. A success must include evidence_ref. The independent_key, evidence digest, and optional content hash support deduplication. Repeated use of one file cannot masquerade as independent practice.

C. Reuse Admission

Admission identifies the task entity and target version, retrieves candidates, and then checks uniqueness, content completeness, conditions, and parent-chain continuity. Conflicts or budget truncation cause fallback to the exact source; unavailable sources yield unknown. This models recalling context and comparing conditions before using prior experience. Production compose_context supplies candidates only. The research parser and host perform final admission, so retrieval itself must not be described as understanding.

D. Effects of Failure

Failure does not silently delete experience. Its evidence, environment, version, and first answer are retained. Failure of mature experience produces disputed; failure of a candidate remains in its observations and blocks unconditional promotion. Narrowed applicability creates a revision, whereas a wholly unsupported claim can become invalidated. The record preserves why the claim was previously accepted and why it is now revised.

IV. Algorithms and Implementation

A. Algorithm 1: Practice Recording and Candidate Creation

Inputs are goal G, action A, environment snapshot E, observation O, provenance s, and expectation $\hat{O}$. An unfinished action or missing evidence is recorded as unknown and cannot trigger promotion. Success creates or retrieves an experience object and checks independent keys, evidence digests, and content hashes; duplicate evidence is rejected. Failure records a counterexample and changes mature experience to disputed or retains a candidate. Observations, evidence digests, experience versions, and domain-skill updates commit within one BEGIN IMMEDIATE transaction in MemoryStore. ButianService subsequently appends practice_observed to a separate service store. These operations are not a cross-database atomic commit.

$$r_t \in \{\text{success}, \text{failure}, \text{unknown}\} \tag{5}$$

B. Algorithm 2: Beta Updating of Experience Belief

Under the working model that a claim succeeds on the next condition-matching task, let $\theta \sim \text{Beta}(\alpha, \beta)$. Evidence weight is $w = qdm$, where q is evidence quality, d an independence assessment, and m environment match. Success yields $\alpha' = \alpha + w$, $\beta' = \beta$; failure yields $\alpha' = \alpha$, $\beta' = \beta + w$. The posterior mean is $\mu = \alpha/(\alpha + \beta)$. Its increase after success is $\Delta\mu = w\beta/[(\alpha + \beta)(\alpha + \beta + w)]$. Thus success increases working belief with diminishing marginal change as parameters grow. This characterizes evidence accumulation without proving the claim.

$$\theta \sim \text{Beta}(\alpha, \beta), \mu = \frac{\alpha}{\alpha+\beta} \tag{6}$$

Production record_practice_outcome currently uses bounded adjustments of confidence: failure subtracts 0.1, and qualifying candidate promotion adds 0.1. It does not automatically call BetaBelief. The Beta model is an optional representation supported by the mathematical module and controls; production confidence is not a Beta posterior. Fractional weighting approximates a weighted likelihood and does not solve evidence duplication. Independence assessment requires higher-level deduplication and environmental review. A posterior cannot bypass version admission or practice thresholds, and warm does not mean proved.

C. Algorithm 3: Candidate Selection and Information Gain

For actions that may reduce uncertainty, information gain is $\text{IG} = H(p_{\text{prior}}) - H(p_{\text{posterior}})$; a cost-adjusted score is $\text{IG} - \lambda\text{cost}$. Expected information gain has a mutual-information interpretation only when the posterior distributions and outcome probabilities satisfy normalization and mixture consistency. Butian's expected_information_gain checks explicit posterior dimensions, outcome probabilities, and mixture consistency. It does not infer posteriors automatically from prose.

$$\text{EIG} = H(p) - \sum_y r_y H(p(\cdot\,|y)) \tag{7}$$

D. Algorithm 4: The Role of Thompson Sampling

Thompson sampling allocates exploration among actions or strategies awaiting verification; it is not the experience-formation process. Arm k maintains $\text{Beta}(\alpha_k, \beta_k)$. Each round samples $\tilde{\theta}_k$, selects the largest sample, and updates only the selected arm using Bernoulli feedback. Selection costs $O(K)$ and updating costs $O(1)$. Without state transitions, long-term returns, or value functions, the current component is a context-free Bayesian bandit rather than full Bayesian reinforcement learning.

E. Algorithm 5: Recovery after Restart

Recovery reads committed events, current experience versions, parent chains, and unfinished tasks before reconstructing candidate context. A prepared event without committed status cannot establish successful external action. Unconfirmed tool termination remains unknown. A version mismatch permits only candidate use and renewed evidence acquisition. Recovery continues the practical process rather than merely reinserting old text into context.

F. Pseudocode with Inputs, Outputs, and Error Branches

The responsibility boundaries below are explicit. Algorithm 1a is the production practice interface; Algorithms 2a–4a are mathematical modules. Algorithm 5a describes host-side recovery and admission rather than a single implemented function. Algorithm 6 is the actual main-experiment path and writes warm directly; it does not replace candidate validation through Algorithm 1a. Numbers indicate execution order; blocks terminate with END.

Algorithm 1a. Record Practice Feedback (Production Interface)

text 01 INPUT: member, memory_id, content, outcome, independent_key, evidence_ref 02 OUTPUT: memory_id, status, successes, failures; or an explicit error 03 REQUIRE outcome IN {success, failure, unknown}; independent_key nonempty 04 IF success AND no evidence_ref: RAISE practice_success_evidence_required 05 Service checks evidence ownership and content hash; RAISE if invalid 06 BEGIN IMMEDIATE (member-memory transaction) 07 IF independent key or evidence reference/digest already used: RAISE duplicate 08 Read experience; create candidate if absent; check requested content hash 09 Insert observation, environment, digest; count successes s and failures f 10 IF failure: mature -> disputed; otherwise retain candidate; reduce scores 11 ELSE IF success AND s>=2 AND f=0 AND old state IN {candidate,pending_validation}: set warm and increase bounded scores 12 ELSE: no promotion; unknown contributes neither success nor failure 13 Save revision, parent chain, provenance, domain statistics; COMMIT 14 ON transaction exception: ROLLBACK; rethrow 15 Append practice_observed separately; do not claim two-store rollback 16 RETURN actually committed state and counts

Algorithm 2a. Beta Evidence Summary (Optional Mathematical Module)

text 01 INPUT: alpha>0, beta>0, outcome, q,d,m IN [0,1] 02 OUTPUT: updated parameters and mean; production lifecycle unchanged 03 Check finite parameters and valid ranges; RAISE if invalid 04 w <- q*d*m 05 IF success: alpha <- alpha+w 06 ELSE IF failure: beta <- beta+w 07 ELSE IF unknown: leave parameters unchanged 08 ELSE: RAISE invalid_outcome 09 RETURN alpha, beta, alpha/(alpha+beta) 10 NOTE: Algorithm 1a deduplicates; fractional weight proves no independence

Algorithm 3a. Rank Verification Actions (Explicit Posterior Control)

```text 01 INPUT: prior p; outcome probabilities r; posteriors p_y; cost; lambda 02 OUTPUT: EIG and cost-adjusted scores for valid actions 03 FOR EACH action a:
04 Check finite nonnegative p,r,p_y; matching dimensions; normalization
05 Check sum_y r_y*p_y[i] matches p[i] within tolerance for each i
06 RAISE if invalid; do not guess posteriors from text
07 eig[a] <- H(p)-sum_y r_y*H(p_y)
08 score[a] <- eig[a]-lambda*cost[a]
09 END FOR 10 RETURN score-ranked actions; the caller decides execution ```

### Algorithm 4a. Thompson Exploration Allocation (Bandit Control)

text 01 INPUT: positive alpha[k],beta[k] for K arms; random-number generator 02 OUTPUT: selected arm; its parameters updated after feedback 03 REQUIRE K>0; finite positive parameters 04 FOR k=1 TO K: sample[k] <- BetaSample(alpha[k],beta[k]) 05 END FOR 06 selected <- ARGMAX sample[k], with implementation-defined tie handling 07 Execute selected; await verifiable Bernoulli reward 08 IF unknown: fabricate no reward and perform no update 09 ELSE IF reward=1: alpha[selected] <- alpha[selected]+1 10 ELSE IF reward=0: beta[selected] <- beta[selected]+1 11 ELSE: RAISE invalid_reward 12 RETURN selection and parameters; no automatic experience promotion

### Algorithm 5a. Conditional Admission after Recovery (Host Workflow)

```text 01 INPUT: persisted tasks, events, experience, chains, current conditions 02 OUTPUT: auditable candidate context and outcome status; no automatic success 03 Open service and member stores; read current versions and provenance 04 Check parent chains; report corruption without inventing missing versions 05 FOR EACH unfinished action:
06 Recover committed outcomes only with verifiable evidence
07 Retain unknown for uncommitted prepared actions, unconfirmed exit,
or missing evidence 08 END FOR 09 Retrieve candidates; admit by entity, version, completeness, and conflict 10 IF inapplicable: reacquire source; IF unavailable: RETURN unknown 11 RETURN candidates, sources, unresolved matters; host must practice again 12 NOTE: experiments test normal close/open, not arbitrary power failures ```

### Algorithm 6. Multi-round Rule-Reuse Benchmark (Executed Research Path)

text 01 INPUT: shared task stream, exact source archive, method, memory directory 02 OUTPUT: first answers; correct/wrong/unknown; read/query/write counts 03 IF memory enabled: compose_context; parse a unique entity rule 04 IF version admission enabled: reject mismatch; ELSE permit old rules 05 IF no usable rule AND source_available: scan or read exact index 06 answer <- fixed solve(rule,task); freeze first answer 07 Score against reference; later correction cannot change first score 08 IF first answer wrong AND source_available: reacquire correct rule 09 IF source acquired AND writes allowed: append_event; MemoryStore.put(status=warm,confidence=1) 10 At round 8 close/open; no-persist switches to an empty directory 11 RETURN round record; repeat for subsequent rounds 12 NOTE: no candidate validation, automatic induction, or disputed-state study

### G. Dream Consolidation, Hypothesis Verification, and Subsequent Practice

Dreaming is offline organization of existing practice material. DreamRunner.run creates a source snapshot when

material volume, new-material, interval, and budget thresholds are met. The host supplies distilled summaries, abstractions, and hypotheses; the system does not derive rules automatically from text. distill preserves source references, abstract retains support, counterexamples, and conditions, and propose checks continuing source validity. Verification must pass schedule and start before supported or rejected outcomes are accepted, and evidence is checked by the service. supported creates only a candidate; later successful practice is still required for warm. Verification separates testable hypotheses from unchecked generalization without guaranteeing correctness on arbitrary tasks.

### Algorithm 7. Dream Formation and Reuse after Verification

text 01 INPUT: practice material, host-proposed rule, verification budget, later tasks 02 OUTPUT: supported/rejected/unknown; candidate; later practice records 03 Check new material, interval, budget; skip consolidation if unmet 04 Freeze sources; host distills, abstracts conditions, and proposes hypothesis 05 Schedule verification; start only if authorized and sources remain valid 06 Execute probes distinct from training; save observations and evidence 07 IF counterexample: record rejected; use source fallback on later tasks 08 ELSE IF insufficient evidence: record unknown; do not claim validation 09 ELSE: record supported and create candidate 10 Freeze first outcomes and record independent practice feedback during reuse 11 Promote to warm only at production threshold; preserve failure evidence 12 RETURN verification/reuse ledger; supported does not mean universal validity

### H. Conditional Propositions for the Bounded Loop

Proposition 1: If every state update uses the MemoryStore practice interface, candidate promotion requires at least two success observations passing interface deduplication and no failures. The promotion branch establishes this by checking counts transactionally and requiring the prior state to be candidate or pending_validation. This proves a program gate, not external evidence truth, and it does not cover manual writes bypassing the interface. Two successes are necessary at this interface, not sufficient for general validity.

Proposition 2: If the current task version is identifiable and the host strictly enforces version admission, an old-version candidate cannot directly yield an accepted current-version result. Matching candidates proceed normally; mismatches require new evidence or unknown. The claim depends on actual version identification and enforcement. The no-version ablation removes that assumption; its errors demonstrate the guard's role rather than universal immunity to errors.

Proposition 3: An unknown observation contributes neither a success nor a failure. Counting includes only the corresponding explicit outcomes. Unknown observations can retain records and versions but cannot increase support. This prevents missing sources or unconfirmed termination from being interpreted as success. Unknown outcomes still reduce completion and remain in the evaluation denominator; they are the cost of honestly representing uncertainty.

A complete process also requires host execution, reliable observations, and later reuse. Database success establishes persistence only. Validity across tasks requires external outcomes, and long-term benefits require cross-round controls. Executable closure and epistemic correctness are therefore argued separately: interfaces, transactions, and recovery controls support the former; the stated task families bound the latter.

### I. Complexity and Implementation Boundaries

Practice writes involve a fixed set of table-operation types, but success/failure counting and domain aggregation grow with observations. For $N$ relevant observations and $N_d$ domain observations, conservative aggregation cost is $O(N + N_d)$ without stronger index guarantees. The entire feedback operation is not $O(1)$, and a fixed number of SQL statements does not make their runtime constant. Current OR-LIKE retrieval costs conservatively $O(ML)$ for $M$ experiences of average content length $L$. Candidate parsing costs $O(B)$ for $B$ context characters; version-chain checking costs $O(V)$; Beta updating is $O(1)$; Thompson selection is $O(K)$. Pareto budget allocation depends on frontier size, capped at 100000. These are program costs, not host-token costs or human cognitive complexity.

## V. Experimental Design

### A. Main Experiment and Scope

The main experiment is a rule-reuse benchmark, not a test of the full experience-formation process. Its remember function calls MemoryStore.put directly with warm and confidence 1.0, bypassing candidate thresholds, independent practice validation, and disputed-state handling. A generator supplies entities and versions; a fixed solver applies rules; an evaluator compares references; source rules are then cached for later reuse. No language model is called. The experiment measures infrastructure supporting a practical cycle rather than language generation. Ten seeds, 12 entities, 12 rounds, and seven conditions produce 10080 executions. Version drift occurs in rounds 5 and 9, source outages in rounds 6 and 7, and normal service close/reopen in round 8.

### B. Conditions and Controls

B0-scan scans the archive every round; B0-index uses an exact version index. B1-static writes only in the first round. B2-full enables retrieval, version checking, source fallback, feedback writes, and persistence. no-version removes admission, no-write disables experience writes, and no-persist uses an empty directory after restart. First outcomes are frozen before feedback, and correct, wrong, and unknown are mutually exclusive. The statistical units are the 10 seeds, not repeated rounds treated as independent samples.

### C. Metrics and Hypotheses

Primary measures are first-attempt completion, version errors, unknown outcomes, and source reads. Secondary measures include context queries, writes, logical operations, local runtime, and version-chain audits. H1 predicts fewer repeated source reads and avoidance of obsolete rules under drift and outages with version admission. H2 predicts retained reusable experience after restart. H3 allows increased local overhead, so fewer source reads need not imply lower end-to-end latency.

### D. Lifecycle Interface Control and Limitations

A separate 24-directory control uses record_practice_outcome: first success creates a candidate; the same evidence with a new independent key is rejected; a second distinct evidence item promotes to warm; failure produces

disputed; the script manually revises to candidate; normal close/open is followed by checks of content, parent chains, and retrieval. This verifies state guards, not autonomous host induction, true evidence independence, or claim correctness. Evidence and revisions are scripted. Twenty-four repetitions are not 24 scientific discoveries. This control complements rule reuse but was originally not one end-to-end experiment jointly measuring formation and task benefit; Section V.G closes this gap.

### E. Paired Replay of Dream Effects

Ten seeds generate linear and saturating sensors, yielding 20 material groups. Initial observations are at x=1,2,3. The current host proposes an affine candidate based on equal adjacent differences; Python replays this fixed induction template through production dream interfaces. The no-dream condition reads sources for each problem. Unverified reuse adopts the candidate directly. Verified reuse probes x=5,10, rejects and falls back on a counterexample, or saves a candidate and records subsequent feedback. Tests at x=4,7,9,12 repeat over four rounds, giving 320 tasks per condition. Probe and test inputs differ. Each condition has identical initial material and 60 training reads. Saturating sensors plateau above x=6, so the initial three points cannot distinguish the families. This examines overgeneralization and verification cost; it is not a blinded autonomous-discovery study and supplies no independent model samples.

### F. Independent Algorithm Controls

A separate two-arm Bernoulli experiment runs 10 fixed seeds and 1000 rounds per seed with reward probabilities 0.30 and 0.70, comparing Thompson sampling, epsilon-greedy, and uniform random selection. It tests exploration allocation only, has actual_model_runs=0, and is neither the main practical experiment nor a full Bayesian-RL study.

### G. End-to-End Gate Experiment Design

To connect formation and task benefit within a single experiment, we add an end-to-end gate benchmark (script run_endtoend_gate_study.py; the criteria were written into a protocol and frozen before the run). It reproduces the main study's task stream statement for statement, so tasks, drift, outages, and restart are identical; the only variation is how experience is admitted as reusable. Six conditions: G0 exact indexing; G1 writes a source rule as warm the first time it is fetched, which is the B2-full write path of Table 1; G2 records each source read as one independent practice and requires two mutually distinct evidence items before promotion; G3 adds, on top of G2, a bad claim whose state is candidate; G4 equals G3 except that candidate-state claims are also treated as reusable; G5 adds, on top of G2, a bad claim carrying two independent practice records. Thus G3 minus G4 isolates the state gate, G3 minus G5 isolates the second independent practice, G2 minus G1 measures the total cost of the gate, and G2 minus G0 measures the benefit of experience against no experience.

Bad claims are injected before rounds 6 and 10 into the entity that drifted in the preceding round: the content is the pre-drift old rule and the version field is rewritten to the current version, so version and content are internally consistent while the content is stale. This is the most common production form of a bad claim and the only external shock the experiment imposes. Twenty seeds (the original ten plus ten new), 12 rounds, and 12 entities give 17,280 executions. The replication self-check requires G1 to match the published B2-full row by row on the ten original seeds and repeated runs on the same machine to match field by field; the statistical unit is the seed and intervals are seed-cluster bootstraps.

## VI. Results: Benefits and Costs of Experience Reuse

### A. Main Results

Table 1. First-attempt outcomes and evidence-acquisition costs.

| Condition | First-attempt completion | Wrong | Unknown | Source reads |
|---|---|---|---|---|
| B0-scan | 1375/1440 | 0 | 65 | 42287 |
| B0-index | 1375/1440 | 0 | 65 | 1375 |
| B1-static | 1400/1440 | 0 | 40 | 752 |
| B2-full | 1440/1440 | 0 | 0 | 253 |
| no-version | 1307/1440 | 133 | 0 | — |
| no-write | 1375/1440 | 0 | 65 | 1375 |
| no-persist | 1440/1440 | 0 | 0 | 373 |

B2-full names all switches of the rule-reuse policy, not the full experience lifecycle. Relative to exact indexing, completion improves by 4.51 percentage points, with a seed-cluster bootstrap 95% interval of [3.68,5.35]. Reads fall from 1375 to 253, an 81.60% reduction. Removing version checks produces 133 obsolete-rule errors: version is an applicability condition, not an incidental field. Discarding experience on restart adds 120 reads, showing that persistence changes subsequent evidence-acquisition actions.

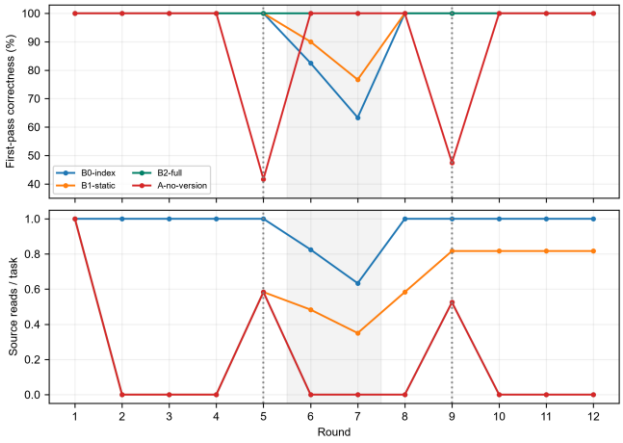



Figure 1. Outcomes under round-specific drift, outages, and restart.

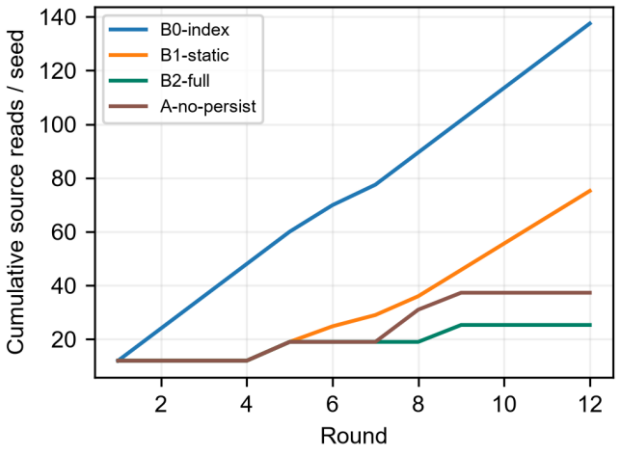



Figure 2. Effect of reuse on cumulative source access.

### B. Round-by-round Attribution

Table 2a. First outcomes by round and B2 external-source reads.

| Round | Index correct/120 | B2 correct/120 | B2 reads |
|---|---|---|---|
| 1 | 120 | 120 | 120 |
| 2–4 (each) | 120 | 120 | 0 |
| 5 (drift) | 120 | 120 | 70 |
| 6 (outage) | 99 | 120 | 0 |
| 7 (outage) | 76 | 120 | 0 |
| 8 (restart) | 120 | 120 | 0 |
| 9 (drift) | 120 | 120 | 63 |
| 10–12 (each) | 120 | 120 | 0 |

Both policies read 120 sources in round 1, providing no cold-start advantage. In rounds 2–4, B2 reuses unchanged rules while indexing still reads 120 sources per round; accuracy is identical. Round 5 updates 70 entity instances: admission rejects 70 old rules and reacquires sources while retaining the other 50. Round 9 requires 63 new reads. Thus total B2 reads decompose exactly as 120+70+63=253; 1122 avoided reads arise from subsequent repeated tasks.

All completion differences occur in rounds 6 and 7. The index returns 21 and 44 unknown outcomes because sources are unavailable; B2 solves these 65 tasks using previously saved version-matching rules. Both solve all 1375 available-source tasks. Among the 65 unavailable-source tasks, index completion is 0/65 and B2 completion 65/65. The 4.51-point gain is therefore an information-availability benefit under this protocol, not stronger reasoning, formulas, or language understanding. Updating rules in round 5 before outages gives B2 a learning opportunity; simultaneous first-time version changes and outages are not covered.

Without admission, 70 round-5 and 63 round-9 tasks use obsolete rules, totaling 133 first-attempt errors. Feedback occurs after freezing the first answer. Available sources then correct the cache, preventing propagation into later stable rounds. After normal restart, B2 has 120 hits, whereas the empty-directory policy rereads 120 sources. Both solve 120/120. Here persistence avoids rereading; it does not improve round-8 accuracy or guarantee power-failure recovery.

### C. Operation Costs and Boundaries

Table 2. Operation ledger for experience formation and reuse.

| Condition | Reads R | Queries Q | Writes W | Logical operations |
|---|---|---|---|---|
| Exact index | 1375 | 0 | 0 | 1375 |
| B2-full | 253 | 1440 | 253 | 1946 |
| B1-static | 752 | 1440 | 120 | 2312 |
| no-persist | 373 | 1440 | 373 | 2186 |

B2-full increases logical operations by 41.53%, averaging 8.44 ms locally versus approximately 0.0102 ms for indexing. The mechanism reduces source acquisition and obsolete-rule errors, not necessarily latency. This local benchmark does not measure human learning cost. With equal operation weights, reuse has no total-operation advantage. Savings may outweigh management overhead only when external reads are sufficiently more expensive than local retrieval and writes. External fees, network latency, and model tokens were not measured, so an application break-even point cannot be established.

### D. Independent Bayesian Bandit Control

Table 3. Exploration-allocation control.

| Policy | Mean total reward | Mean pseudo-regret | Seeds |
|---|---|---|---|
| Thompson | 704.4 | 4.32 | 10 |
| epsilon-greedy | 672.8 | 25.44 | 10 |
| Uniform | 494.3 | 202.60 | 10 |

Thompson sampling concentrates relatively quickly on the higher-reward arm in this stationary two-arm Bernoulli environment. This describes the allocator, not the correctness of experience claims or reinforcement-learning capability in the host model.

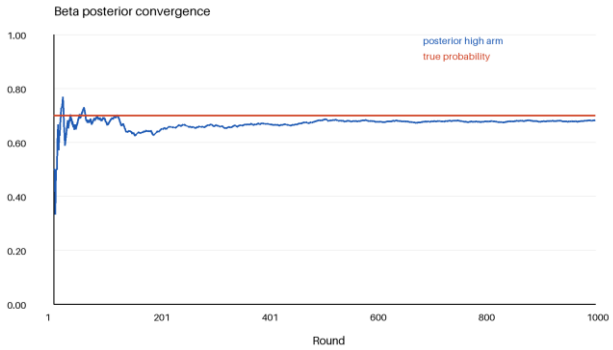



Figure 3. Convergence of the high-reward candidate's Beta posterior mean.

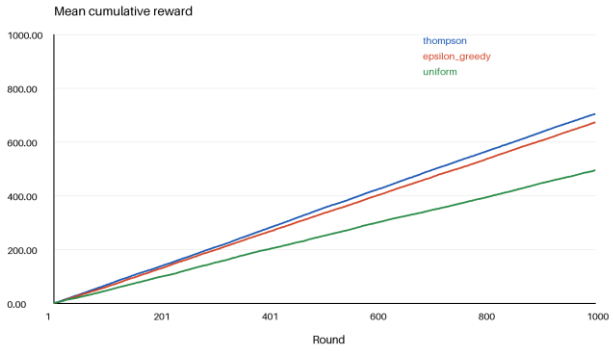



Figure 4. Mean cumulative reward of the three exploration policies.

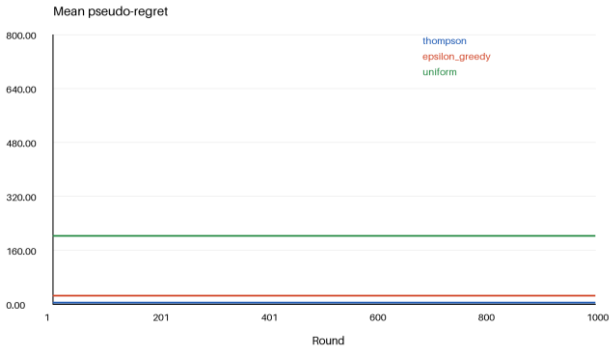



Figure 5. Mean pseudo-regret of the three policies.

### E. Subsequent Dream Benefits and Incorrect Generalization

Table 4. Paired outcomes for candidate verification and later reuse.

| Condition | Correct/320 | Wrong | Later source reads | Verification accesses |
|---|---|---|---|---|
| No dreaming | 320 | 0 | 320 | 0 |
| Unverified reuse | 200 | 120 | 0 | 0 |
| Verified reuse | 320 | 0 | 160 | 40 |

Verification accepts 10 linear candidates and rejects 10 saturating candidates. Linear rules solve 160 later tasks by reuse; the other 160 use fallback. Later reads fall 50%, from 320 to 160. Including 40 verification accesses gives 200, a 37.5% reduction. Including the identical 60 training accesses per condition gives 380 versus 260, a 31.58% reduction. Unverified reuse fails on three above-threshold inputs per saturating sensor per round: 10×4×3=120 errors. Compression into a rule can lose boundaries. Verified reuse matches no-dream accuracy; its benefit is reduced acquisition rather than improved reasoning accuracy.

Accepted candidates pass through the real record_practice_outcome: the first success retains candidate, and the second promotes to warm. This connects dream

candidates, tool verification, and later practice. However, reference outcomes come from the same synthetic generator and remain visible to the evaluator; they are not free external evidence in deployment. The induction template was specified by the host, with zero model calls during script execution. Repeated rounds within a seed are not independent samples. The results establish verification's role in this protocol, not the success rate, token cost, or cross-domain generalization of unconstrained host induction.

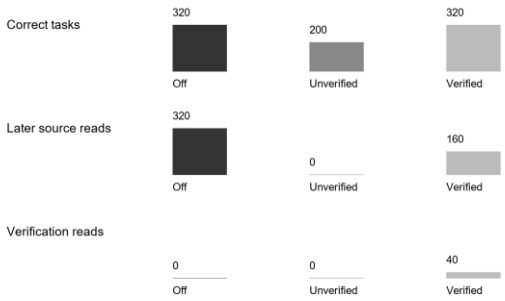



Figure 6. Correct tasks, later source reads, and verification accesses under the three dream conditions.

### F. End-to-End Gate Experiment: Linking Formation to Task Benefit

Table 5. First-attempt outcomes and gate costs of the six conditions (20 seeds pooled, 2880 tasks per condition).

| Condition | Correct | Errors | Unknown | Source reads | Writes | Gate rejections |
|---|---|---|---|---|---|---|
| G0 exact index | 2759 | 0 | 121 | 2759 | 0 | 0 |
| G1 direct write to warm | 2880 | 0 | 0 | 496 | 496 | 0 |
| G2 production gate | 2880 | 0 | 0 | 736 | 736 | 240 |
| G3 gate plus candidate bad claim | 2844 | 0 | 36 | 992 | 992 | 449 |
| G4 no gate, candidates reusable | 2653 | 209 | 18 | 752 | 943 | 0 |
| G5 gate plus two-evidence bad claim | 2634 | 209 | 37 | 1601 | 1792 | 868 |

The replication self-check yields two results: on the ten original seeds G1 totals 1440 correct answers and 253 source reads, identical row by row to B2-full in Table 1; and the task stream matches the published archive item by item, while a repeated run on the same machine matches all 864 rows field by field. The 4.51 percentage points of Table 1 are therefore obtained under the shortcut of writing a fetched source rule to warm immediately, which bypasses the candidate gate and the second independent practice; with the gate returned to the loop, the gain of G2 over G0 is 4.20 percentage points (6.05 additional solved tasks per seed, seed-cluster bootstrap 95% interval [3.65, 4.79] percentage points, positive for all 20 seeds). The two numbers are of similar magnitude but different meaning: the former is an upper bound describing the situation in which experience already exists, the latter includes the cost of the gate.

The cost of the gate is an identity: G2 reads 12.0 more sources per seed than G1 (interval [12, 12]), exactly equal to its 12.0 gate rejections per seed, while accuracy is identical (all 20 seeds tied). The gate therefore removes no accuracy; it exchanges each hit that the state does not allow for one source read. Against the experience-free index baseline, G2 reads 2023 fewer sources and solves 121 more tasks, a 73.33% reduction in reads.

The effect of the state gate narrows to one provable point. The same bad claim (256 injections) produces 0 first-attempt errors on G3, 209 errors on G4 with all 256 items fooling the system at least once (median 1 per item), and exactly the same error count on G5 as on G4. Adding the second independent practice therefore does not reduce how often a bad claim deceives: the gate blocks claims that carry only a single evidence impression and have not completed a second independent practice, not claims that carry evidence. The cost of the gate appears as coverage: the 36 unknown outcomes of G3 all fall in rounds 6 and 7, where the gate rejects and no source is available, so a possible wrong answer becomes a missing one.

The feedback channel is one-directional. The probe shows that a warm memory enters disputed after one failure and does not return to warm after two further successes; only an explicit revision reopens it. In the main study all 191 hit-then-wrong-then-corrected events fall inside this one-directional interval (G4 lands in candidate, G5 in disputed). This is not an

implementation error but a direct consequence of the promotion rule requiring a zero failure count, and we record it as an open limitation.

### G. Interpretation

The main study supports condition-matching reuse of known rules, not autonomous formation of claims from practice. B1 caches only the first round and can still read sources after updates; its limitation is repeated acquisition, not inevitable obsolete outputs. Formation, thresholds, and disputes were originally covered only by the interface control; the end-to-end gate experiment of Section VI.F links them to task benefit in one ledger, and the conclusion is that the gate is real but narrow: it blocks claims that have not completed a second independent practice, it does not raise accuracy, and it does not block claims carrying two evidence items. Bandit benefits arise from exploration allocation at a different level. These data cannot be merged into one model-effect number.

### H. Real-Host Multiround Experiment

The preceding three experiments were hosted either by a deterministic solver or by a single-round, non-blinded pilot. To add a real model driving real tools over multiple rounds with objective scoring, a pre-registered protocol was written before any host answer: the model of the current session calls the recall and source-read tools directly over JSON-RPC on standard input and output, with only three allowed actions — recall experience, read the source, and submit an answer. Experience writeback is performed by the harness after exact-match scoring, so the host has no channel for writing experience. Twelve entities are arranged as 3 families x 4; the first two of each family keep one version across all three rounds and the last two are upgraded each round, giving 36 answers in total. This is a pilot and no p-values are reported.

All five pre-registered criteria hold. Stable entities score 6/6 in the first and in the last round, so accuracy does not degrade; drifting entities produce zero obsolete-rule errors in the last two rounds, while each of those rounds reads the source exactly six times, equal to the number of drifting entities; stable entities drop to zero source reads in the last two rounds without losing correct answers; at the last round the six stable memories are warm (three independent practices each, under three independent practice keys), while the eighteen drifting memories split into three per-version entries that all stay candidate; and all 36 answers carry a recall record, with every reuse answer naming a memory that appears in that round's recall payload. Source reads total 24, a 33.3% reduction against the 36 reads that reading for every entity in every round would require, and the saving falls entirely on the stable entities in the last two rounds.

Three limitations must be stated together. First, accuracy is saturated (36/36), so "no degradation" holds as a tie rather than as an improvement and the informative content is behavioural: version admission was executed by the host itself after comparing the version field inside the recalled content, and a host that lazily reused the stale rule would have made six more errors in each of the last two rounds. Second, the host is non-blinded and the arithmetic of each question was executed by a local script; the experiment shows that the channel, version-admission, and reuse chain works when driven by a real host, not that model reasoning improved, and it does not generalize across models. Third, two construction defects that would have made the questions unanswerable were fixed before any answer was produced: rules were originally generated per round, so the archive carried three different RULE lines for the same entity and version, and the archive did not state the conversion formula. With those fixed, a failed criterion was preferred to a claim built on an unanswerable apparatus. The experiment also exposed an interface shortcoming: recall performs substring matching only, with no structured filter by entity or tag, so with limit = 10 only ten of twelve memories are returned and the queried entity itself can be the one missing; the host must raise the limit and filter by itself. This is recorded as an interface defect to fix, not as a mechanism result (the interface was fixed on 2026-11-02; the post-fix re-measurement is the closing paragraph of Section VI.I).

### I. Real-Host Stress Arm (Near-Neighbour Versions and Upstream Bad Experience)

All 36 answers of the previous subsection were correct, and that saturation removes discriminatory power: the memories written back already carry the scored ground truth, so correct reuse only shows that the value was copied; and for drifting entities the version number is embedded in field and endpoint names, so the version is visible at a glance. To make the apparatus discriminating, a second pre-registered protocol (likewise fixed before any host answer) adds exactly two stress axes and keeps every other convention identical. First, near-neighbour version decoys: adjacent versions of the same entity keep identical field and endpoint names and differ only by 3% in numeric values (offset increases by 3 per version, while divisor and factor take 100/103/106/109), so the version can only be resolved from the version field of a RULE line. Second, upstream bad experience: for three entities the memory is already warm before the host ever recalls it, holds a wrong value (offset by 7 or 37), and carries two independent success observations, simulating a previous session that wrote an extraction error into the experience store. The scale grows to 18 entities, four rounds, and 72 answers.

All five pre-registered criteria hold. The clean-stable group answers 24 of 24 correctly with zero source reads in the last three rounds; the nine drifting entities that include the decoys read the source nine times per round in the last three rounds with zero obsolete-version errors, which shows that embedding the version in field names was not what kept the earlier arm free of obsolete-version errors — what actually acts is an explicit comparison of the version field. The three bad-experience entities are all trusted blindly in the first round and scored wrong (3/3); immediately after the failure their status drops from warm to disputed, the host switches to reading the source in the next round, all three answers become correct, and the recovery round is 1. Source reads total 51 against 72 for read-everything, a 29.2% saving, but the 21 saved reads must be decomposed: 18 come from clean reuse, and 3 are bought by trusting the bad experience (the price is three wrong answers), while the bad experience costs nine additional reads over the following three rounds. Reporting 29.2% alone would read a failure as a benefit.

The most informative reading lies in the terminal memory state: the three corrupted memories each end with five success and one failure observation, remain disputed, and their content has not changed by a single character. Memory content is written only at creation and a success observation only moves status and confidence, so these three entities lose reuse value permanently. The boundary of the mechanism can therefore be stated precisely: the chain of scoring, failure observation, demotion and strategy change works (it stops the bleeding), but

there is no content-level conflict detection or repair channel (it does not heal); changing the content requires a new memory record or the dreaming and manual-revision channels, which this protocol does not open. The limitations must be stated together with the result: the host is non-blind and knows that this arm contains a bad-experience group, the bad experience is injected by the harness with known offsets, only three of 72 answers are wrong, and the arm remains a pilot; arithmetic is executed by a local script, so the arm supports claims about the channel and the policy chain, not about model reasoning.

Both gaps were fixed the same day, and the fix was re-measured on the same artefacts. Recall now weights every query token by its document frequency inside that member's own memory, and only a token that is at least four characters long and rare in the store (document frequency at most one quarter of the memory count) acts as a hard gate; the remaining tokens only order the results, and every returned item carries three auditable fields (match_score, matched_tokens, gated_by). Replayed against the 45 real memories of this arm with 72 queries (4 rounds times 18 entities), the median number of returned items falls from 45 (the whole store) to 2.5 (1 for a stable entity, 4 for a drifting entity, whose version is still the host's call), items unrelated to the queried entity fall from 3060 to 0, precision rises from 5.56% to 100%, and no query returns an empty set. The gate only fires when the query really contains a rare long token; asking about an entity the store has never seen falls back to plain OR matching, preferring a neighbouring lesson over pretending there is no experience. For content correction, butian_memory_revise requires evidence this memory has not already used, marks the old version superseded, records content_revision with both content hashes, and restarts the success and failure counters from the revision, so "correct, re-verify, back to warm" is reachable; a regression test pins the sequence in which one failure turns the memory disputed, three further successes still do not unlock it, and an explicit revision plus two independent successes restore warm. Two caveats must accompany these numbers: they are re-measurements of an interface fix, while this arm itself ran before the fix and therefore only observed the old behaviour; and an explicit revision channel is not automatic conflict detection, because the system still cannot tell whether memory content matches its source.

### J. Mis-Attribution, Content Revision and Scale on the Real Host

The first two arms settle "bad experience" and "version admission", but three questions could not be posed before the fixes: what happens when correct content is mis-judged, whether a corrupted memory can be repaired, and whether recall still holds when the store holds hundreds of memories. A third pre-registered protocol (again fixed before any host answer) answers them with 3 groups x 3 families = 9 entities x 4 rounds = 36 answers, on top of 300 same-family but unrelated memories preloaded into the store. Of the 36 answers, 33 are correct, 3 are wrong and none are unknown; all three errors are the designed sentence — the repair group trusting bad experience in the first round.

The three memories of the mis-attribution group hold correct content and were merely marked disputed by a single failure observation. The host reads the source and answers correctly on all twelve questions across four rounds: mis-attribution never makes an answer wrong, but it removes the reuse benefit entirely. The harder reading is the state: every round records one more success observation after scoring, so the cumulative success count climbs from 3 to 6 while the status never leaves disputed, because returning to warm requires at least two successes and zero failures and that one failure stays inside the counting window. Mis-attribution is therefore not an accuracy problem but a cost problem plus a one-way lock.

The three memories of the repair group hold wrong values and sit at warm with two independent success observations. In the first round the host reuses them directly and all three answers are wrong; after the failure the harness applies a content revision supported by a new measurement, the status moves from disputed to candidate, and the counting window restarts at the revision. In the two candidate rounds the host keeps reading the source, answers correctly and earns one success each; the second success turns the memory warm in that same round, and in the fourth round the host stops reading and answers correctly again. The price of the repair channel can thus be measured exactly: one revision, two independent successes, and two further rounds that still read the source — not "trustworthy the moment it is edited".

The scale group adds three hundred same-family but unrelated memories (one hundred per family, all warm with complete content) to the nine task memories. Each of the twelve recalls returns exactly one item with zero unrelated entries, and the host answers all four rounds correctly without reading the source in any round; the same measurement read 45 items and 3060 unrelated entries per query in the previous arm. One boundary must be recorded: when the queried entity has no memory at all, the rare-token gate does not fire and recall falls back to plain OR matching — in the pilot the three unseen entities each pulled back one hundred same-family memories, which is the intended "better one extra neighbour than pretending there is no experience" behaviour, so the formal run seeds one warm memory per scale entity.

Four limitations accompany the result: the host is non-blind and knows the three constructions; this is a single host, a single run, three entities per group, reported as behaviour with no p-values; the mis-attribution, the wrong values and the 300 unrelated memories are all injected by the harness rather than sampled from real incidents; and the recovery rhythm is determined by the rule "at least two successes and no failure means warm", which is a design choice rather than a universal constant. One case remains unmeasured: when the evidence itself is wrong, a revision writes the wrong value in harder, and this arm does not cover it.

### K. Continuous Drift and the Version-Metadata Ablation

The first three real-host arms were close to the accuracy ceiling, and Section VII.D states accordingly that they yield behavioural readings rather than an accuracy difference. An experiment in which everything is answered correctly cannot rule out the possibility that removing the mechanism would also answer everything correctly. The fourth protocol, likewise frozen before the host saw any recall payload, produces the first accuracy difference on the real-host line: 2 groups x 2 arms x 3 families = 12 entities x 4 rounds = 48 answers. Both arms share the same store, the same memories (each pre-seeded to warm through two independent successes), and the same host consumption rule; the only difference is the recall presentation layer. The gated arm returns memory_id, status, and content carrying an explicit version field. The naive arm returns content only, with the version field stripped, modelling a text-only

memory module that injects whatever it retrieves into the context. In the drift group the source document advances one version per round while the memory stays at v1; the stable group keeps v1 for all four rounds as an anti-false-green control. No writeback and no practice observations are recorded in this arm, isolating drift as the single factor.

The difference is 75 percentage points. Under drift the gated arm answers 12 of 12 correctly while the naive arm answers 3 of 12 (25%), with 9 errors; without drift both arms answer 12 of 12, identically. The difference appears in the first drifting round and reproduces in every subsequent round. All four pre-registered criteria hold. The failure mode is pinned separately: each of the 9 wrong answers is exactly the answer computed from the v1 rule (stale_match 9/9), and the harness additionally checks that for every drifting entity the v1 answer differs from the current truth in rounds 2 to 4, so this is reuse of an obsolete version rather than random error. The naive arm never reads the source in any round: with neither version nor status in the presentation, the consumer has no signal on which to base doubt. The gated arm falls back to the source whenever the version disagrees, at a cost of 9 reads. The final memory check shows that drift does not launder the old memory: for all three gated drifting entities the memory remains version 1 and warm across all four rounds.

This reading moves version and status metadata from a design requirement to a component whose removal measurably reduces accuracy. The scope must nevertheless be fixed: the naive arm's behaviour of trusting content when no metadata is present is a modelling assumption, not a measurement of any particular open-source memory module, so the result reads as "version metadata is load-bearing for a consumer that trusts retrieved content", not as "any naive memory system will fail". The stable control excludes the alternative explanation that the naive arm is simply broken, but it does not exclude other viable consumer-side strategies. Because nothing is written back, this arm answers only whether an obsolete version is reused, not whether a new version can be learned. Arithmetic is still executed by a local script; the run is single-seed, single-run, and non-blinded; and 12 independent rules cannot be treated as 48 independent samples, so no p-values are reported. Cross-model consistency is addressed by the external-model replication in the next subsection.

### L. The Same Ablation with External Models: Cross-Model Replication

The host in the preceding subsection was still the model of this session, which is also the designer, so the possibility that the difference is one host's behavioural habit could not be excluded. The fifth protocol, likewise frozen before any external model saw any recall payload, replaces the host with external API models while keeping tasks, rules, memories, recall presentation, and consumption rule identical; only the host changes. Two models answer 48 questions each, 96 in total. DeepSeek-V4-Pro is from a different family than the session host and carries the cross-model claim; DeepSeek-V4.1-Flash is from the same family and serves only as an independent-process replication. The external models do not know the group construction, do not know the hypothesis, and took no part in the design, so the non-blinded-host threat of the previous arm is removed here.

Both external models reproduce the previous arm cell by cell: under drift the gated arm is 12 of 12 and the naive arm 3 of 12; without drift both arms are 12 of 12; source reads are 9 and 0 respectively — identical to the session host in all four cells. All four pre-registered criteria hold on both models, giving the verdict v5_cross_model_replicated. The two-arm gap over the three drifting rounds is 100 percentage points on both, and the nine wrong naive answers are still exactly the obsolete-rule answers. A secondary scoring of the models' own arithmetic gives 39 of 48 on both, cell by cell identical to the primary scoring: all nine errors come from selecting the wrong rule version, and not one comes from choosing the right rule and miscalculating, so arithmetic ability is not the source of the difference. Each model made 57 API calls (48 decisions plus 9 post-source answers), with zero JSON parse failures and zero protocol errors; every request and reply is persisted for audit.

The protocol also fixed an apparatus defect in advance and pre-registered a clean subgroup: the naive presentation strips only the explicit version field and memory_id/status, not the incidental version cues embedded in content (the /api/v1/ endpoint and limit_v1 parameter in the tool family, the f1 and h1 field names in the schema family). The calibration family carries only two plain numbers, scale and offset, and is the only subgroup with no version character at all. The measured two-arm gap in that subgroup is likewise 100 percentage points on both models, and the per-family decomposition is identical. This narrows the conclusion from "the model may have recognised v1 inside the endpoint string" to: even when the content contains no version character whatsoever, removing the metadata still makes the model reuse the obsolete rule. The scope remains fixed: both external models come from one vendor and cannot be extrapolated to arbitrary models; the consumption rule is still a modelling assumption written into the prompt; designer non-blinding remains; and each host is a single run at temperature zero without repeated sampling, so no p-values are reported.

### M. With and Without the Library: Effectiveness on Two Axes

The five preceding subsections all ask whether some component of Butian is load-bearing, and in every arm both sides still use the experience library, so they can establish that a component is necessary but not that the whole is better than doing without it: once accuracy saturates, the drop caused by removing a component only shows that the component matters. The sixth protocol, likewise frozen before any external model saw a single recall payload, changes the comparison. It pits two real workflows against each other: the lib arm queries the library first and reads the source only when the memory is not trustworthy; the noref arm has no memory layer at all and can only read the source. The corpus is a synthetic device manual of 327 lines and 38,063 bytes covering three families (calibration, schema, tool) with 40 entities each, 120 entities in total and one to four version rows per entity. The task set takes eight questions per family, 24 in all, and the remaining 96 entities serve as decoys, so neither arm knows which entities will be asked about. The library is pre-loaded with all 120 entries, each backed by two independent evidence items in the warm state, so a recall may return a task entity, a decoy, or an obsolete version. Drift has three levels: the first round is all v1; the second round revises eight entities; the third keeps those eight at the new value and revises four more for the first time, giving staleness ratios of 0/24, 8/24 and 12/24. Revisions force a redraw of the parameters so that carrying over an obsolete version never coincides with the correct answer. Both arms share the corpus, the task order and the prompt skeleton, run at temperature zero,

and use an independent context per question with no cross-question cache.

The first axis measures cost. With unlimited reads over three rounds and 72 questions, the lib and noref arms score 98.6% and 100% for DeepSeek-V4-Pro and 100% and 100% for DeepSeek-V4.1-Flash. The lib arm makes 20 source reads against 72 for noref, a 72.2% saving, and prompt tokens fall from 1,054,132 to 335,918 (31.9%) on the Pro host and from 1,021,611 to 331,183 (32.4%) on Flash. The two hosts make exactly the same number of reads because whether a read is needed is fixed by the frozen drift schedule rather than by model preference. The only miss on the Pro host is one third-round question where the source was read but the answer could not be parsed and was scored unknown; that is output-format jitter, not a defect of the memory layer.

The second axis measures budget. With only 6 or 12 source reads per arm per round and 24 questions, the lib arm completes 18/24 and 24/24 while noref completes only 6/24 and 12/24, a gap of 50 percentage points that is identical on both hosts. The noref readings map one-to-one onto its budget: each read answers exactly one question, so the arm stops working when the budget runs out. The lib arm spends its budget only on the questions whose memory is no longer trustworthy; 6 reads exactly cover the six stale entities and 12 cover all twelve. Treating the read decision as a binary gate and auditing it question by question: the stale counts in the three rounds are 0, 8 and 12, the lib arm reads 0, 8 and 12 times, and the read sets coincide with the stale sets exactly, with zero wasted and zero missed reads and 24 of 24 on both hosts.

The failure mode deserves a paragraph of its own. On the questions where the second-axis budget is exhausted and the memory is obsolete, the lib arm returns unknown every time and never guesses from the old value; the noref arm likewise returns unknown rather than inventing a number once its budget is gone. The noref arm therefore loses not by being wrong but by being unable to proceed: 25% and 50% are the shares of work it can finish, and the rest is an explicit refusal. This is consistent with the requirement to report an error rather than give a wrong answer, and it makes the meaning of the improvement honest: what the library buys is more work finished at the same budget (threefold and twofold), not wrong answers turned into right ones.

The scope must be fixed. The manual is synthetic, with single-valued and unambiguous entries, so the ambiguity, conflicts and missing pages of real documents are not covered. The staleness marker is computed by Butian itself, and this arm does not inject cases where the marker is wrong; if the marker is unreliable, the 24-of-24 gate degrades into misreads or missed reads. Independent contexts overestimate the source-reading cost of noref, and that bias works against noref. Each level has only 24 questions: the 50-point gap is far above noise, but the absolute numbers depend on corpus construction and cannot be extrapolated to a percentage on arbitrary tasks. The arm measures consumption only, not write-back, not multi-turn reference, and not memory contamination; each recall returns exactly one entry, which is the ideal case at a limit of 20. Both hosts come from one vendor, so the run covers one same-family and one cross-family host and nothing more.

### N. Building the Library Itself: Effectiveness on the Formation Side

The six preceding subsections all pre-populate the library: 120 entries, two independent pieces of evidence each, status warm, with the model only consuming them. That shows a populated library beats no library, but not that a library the host builds for itself is still worth having, which is what a real deployment looks like. The seventh protocol (also frozen before any host saw any recall) empties the library: the form arm (recall plus on-demand source reads plus practice and revise write-backs) starts from nothing and builds as it goes, while the noref arm only reads the source. Corpus, tasks and drift schedule are identical to the sixth study, with a single host, DeepSeek-V4.1-Flash, temperature zero and independent context per question.

Both arms answer 72 questions. The form arm scores 72 of 72 and the noref arm 72 of 72, so accuracy does not drop. Source reads fall from 72 to 36, a 50 percent saving; restricted to the second and third rounds, after the library exists, they fall from 48 to 12, a 75 percent saving. Prompt tokens fall from 1,021,179 to 604,974, or 59.2 percent of the opponent. The decision split is clean across rounds: all 24 first-round questions read the source and file an entry; in the second round the eight revised entities read the source while the other 16 reuse memory; in the third round the manual has 12 stale entities but eight of them were already re-filed to the new value in round two, so only four are still unknown to the library, giving four reads and 20 reuses. What is saved is exactly the class the library can serve: present, version-matched and usable.

Formation is not free. The host files 24 entries (the independent key comes from the source line), revises 12 (one for each manual revision) and records zero unchanged, skipped or failed write-backs; the lifecycle walks candidate, reuse, and superseded replaced by a new chain. The first round is pure investment, 48 calls against 24, so it is more expensive than the opponent and the payoff only starts in round two; the 50 percent figure is a three-round average and must not be read as an immediate saving. Contamination and stale reuse are both zero: the harness never corrects what the host writes, every stored line is copied verbatim from the source, all 12 version-mismatched questions diverted to the source instead of forcing the old value, and all 36 version-matched questions reused directly, with the gate correct on 48 of 48 questions.

Scope must be stated. There is one host, one seed, 24 entities and three rounds, so nothing generalizes to arbitrary models or documents; protocol section 7.7 explicitly permits reusing a self-filed candidate (reading the same source twice is not independent evidence, and insisting on the default rule of two independent successes would make formation cost more than the benefit), which is a modeling assumption rather than Butian's default behaviour and is bounded by the contamination criterion; the source line for each entity and version pair is registered as an evidence item by the harness and write-backs use the real evidence identifier, which is a harness guarantee and not a model capability; and the corpus remains a synthetic, unambiguous manual without any injected case of the staleness marker itself being wrong.

## VII. Discussion: From Human Practice to Host-Agent Mechanisms

### A. What Is Modeled

The model represents the structure of practice rather than the full human mind. It retains observable goals, actions, outcomes, conditions, evidence, counterexamples, revisions, and renewed practice. Mao Zedong's *On Practice* can inform the design through its account of knowledge arising from and

being tested in practice [7]. Philosophy does not prove program correctness, which instead depends on guards, transactions, and experiments. Long-term-memory research also motivates recovery of task context without equating software fields with human memory [1-2].

### B. Bayesian Components Are Not Experience Itself

Temporal-context models discuss the relationship between retrieval and changing contexts [8]. We use this to motivate environment-aware recovery, not to establish that SQLite structures are human memory. The Thompson-sampling tutorial describes posterior-sampling exploration [9]; our two-arm stationary control covers only its simplest Bernoulli case and cannot establish performance under nonstationarity, partial observability, or delayed real-host feedback.

A Beta posterior describes how evidence changes a belief summary under a working model. It does not establish understanding and does not replace counterexamples, versions, or provenance. It is one numerical component within a lifecycle. Thompson sampling allocates opportunities among verification actions; it is neither the complete formation process nor full reinforcement learning. Allocator performance and lifecycle effectiveness require separate evaluation.

### C. Host AI and Model Training

Runtime experience changes retrievable context, evidence, and policy selection, not parameters. Host AI can inspect candidates, act, and submit feedback, but its free-text responses require separate records of first answers, visible context, tool outcomes, and feedback. The current 15 same-agent, nonblinded exploratory records are not independent model experiments; the 10080 main executions also have no language-model calls. Experience may supply candidate post-training data, but parameter-change claims require fixed pre/post evaluation, forgetting assessment, and rollback evidence. Runtime writeback is not described as self-evolution.

### D. Threats to Validity

Contractual synthetic tasks and deterministic solving reduce model noise but limit external validity. Ten seeds are clusters; 1440 repeated tasks are not 1440 independent samples. Drift and outage schedules are protocol choices, not real-world distributions. Local latency does not equal network or token cost, and same-agent pilots expose previous answers. Interface regression checks verify implementation behavior and are excluded from task-benefit statistics. The end-to-end gate experiment places formation and benefit in one ledger, but its bad claims are a fixed shock imposed by the experimenter rather than an estimate of production frequency; G1 and G2 both reach 100% accuracy on 20 seeds, so the apparatus is insensitive to accuracy and only resolves the two-round outage coverage, the read cost, and first-attempt errors after injection. The real-host experiments face non-blinding directly: the host is also the designer. Across the first three arms only 6 of 144 answers are wrong, and all six are the designed first-attempt uses of injected bad experience, so those arms yield behavioural readings (version admission executed by the host, reuse driving stable-entity reads to zero) rather than a mechanism-level accuracy difference; their arithmetic was executed by a local script, so they cannot be read as evidence about model reasoning either. The fourth arm closes that gap with an ablation: stripping version and status metadata from recall drops accuracy under continuous drift from 12 of 12 to 3 of 12, while the non-drifting control is 12 of 12 in both arms. Two limits remain: the naive arm's trust in unannotated content is a modelling assumption rather than a measurement of a real memory module, and the run is still a single non-blinded host session. The fifth arm replaces the host with two external API models, one from a different family than the session host, and reproduces all four cells exactly with the clean subgroup holding as well, removing the non-blinded-host threat; both external models nevertheless come from one vendor, the consumption rule remains a modelling assumption written into the prompt, and designer non-blinding remains, so no extrapolation to arbitrary models is made. The sixth arm looks not at components but at the workflow itself: the library-equipped arm finishes 72 questions with 20 source reads where its opponent needs 72, at 31.9% of the prompt tokens and with no loss in accuracy; and when each arm is capped at 6 or 12 reads, completion is 75% against 25% and 100% against 50%, identical on both cross-family hosts, with the gate wasting and missing nothing in 24 of 24 decisions and both arms returning unknown rather than guessing once the budget is gone. The untested premise of that arm is equally explicit: the staleness marker is computed by this system, and cases where the marker itself is wrong are not covered. A seventh study empties the library so that the host files and revises its own entries: accuracy stays equal over 72 questions, source reads fall from 72 to 36 and from 48 to 12 once the library exists, prompt tokens reach 59.2 percent of the opponent, 24 filed entries and 12 revisions match the manual revisions one for one, and contamination and stale reuse are both zero; its untested premises are a single host and a single seed, and the permitted reuse of a self-filed candidate is a modeling assumption stated in the protocol. The stress arm pushes the same question one step further: after a bad claim backed by two independent evidence items is injected, the host trusts it in the first round and is scored wrong, then switches to reading the source because of the status demotion, while the corrupted content is never repaired — status-level demotion stops the bleeding but does not heal.

## VIII. Conclusion and Future Work

This paper models experience formation as goal-directed action producing observations, comparison with expectations yielding feedback, provenance and condition checking producing candidates, and repeated practice and counterexamples determining stable or disputed states. Versions and restart recovery preserve continued verification. Butian's database, evidence, version chains, context recovery, and feedback interfaces implement observable stages of this process.

The experiments support five bounded conclusions. First, version-admitted rule reuse reduces source reads by 81.60% against exact indexing, while removal of version admission produces 133 obsolete-rule errors. Second, persistence preserves reuse across restart. Third, Beta updating and Thompson controls follow their intended behavior in synthetic numerical environments. Fourth, with the candidate gate returned to the loop, the end-to-end gate experiment reproduces B2-full exactly (1440 correct answers and 253 reads) on the ten original seeds and quantifies the net effect of the gate on 20 seeds: 121 more tasks solved and 2023 fewer source reads (73.33%) than exact indexing, at the cost of 12 additional reads per seed; the state gate blocks bad claims carrying a single evidence item (0 errors versus 209 errors) but not bad claims carrying two independent practices. In the additional dream replay, verified reuse and no dreaming both solve 320 tasks, while verification-inclusive acquisition falls by 37.5%;

unverified reuse yields 120 overgeneralization errors. This supports verification before reuse in the two studied families, not unrestricted free-text induction benefits. Fifth, with the same process attached to a real host, all five pre-registered criteria hold over 36 answers: stable entities reach zero source reads in the last two rounds, drifting entities make zero obsolete-rule errors, stable memories rise to warm while drifting ones stay candidate per version; but all 36 answers being correct means the experiment yields behavioural readings only, not an accuracy difference. Adding stress axes to the apparatus (near-neighbour version decoys and upstream bad experience, 18 entities, four rounds, 72 answers) leaves the clean group at 24 of 24 with zero reads in the last three rounds and the decoy group at zero obsolete-version errors; the bad-experience group is wrong on all three first-round answers and recovers completely in the next round, yet the corrupted content is never repaired: that memory collects three further successes (ending at five successes and one failure) and stays disputed, showing that the status gate stops the bleeding but does not heal. The fourth arm ablates version and status metadata alone: with the same store, the same memories, and the same host consumption rule, removing version and status from the recall presentation drops accuracy under continuous drift from 12 of 12 to 3 of 12, with all nine wrong answers exactly equal to the obsolete-rule answer, while the non-drifting control is 12 of 12 in both arms. This is the first accuracy difference on the real-host line and turns the metadata from a design requirement into a component that can be removed and tested, although the naive consumer behaviour remains a modelling assumption rather than a measurement of a real memory module. The fifth arm replays the same ablation with two external API models over 96 answers and reproduces the four cells exactly (12 of 12 versus 3 of 12 under drift, 12 of 12 in both arms without drift), with the failure mode still exact reuse of the obsolete version and the same 100-percentage-point gap in the calibration subgroup that contains no version character; secondary scoring of the models' own arithmetic matches the primary scoring cell by cell, excluding arithmetic ability as a confound. The non-blinded-host threat is thereby removed, but both external models come from one vendor and no extrapolation to arbitrary models is made. The sixth arm changes the object of comparison from components to workflows: the library-equipped arm finishes 72 questions with 20 source reads where its opponent needs 72, at 31.9% of the prompt tokens and without losing accuracy; with each arm capped at 6 or 12 reads, completion is 75% against 25% and 100% against 50%, a 50-point gap identical on two cross-family hosts, with zero wasted and zero missed reads in 24 of 24 gate decisions and explicit unknown rather than guessing when the budget runs out. This moves the claim from components being load-bearing to more work finished at the same cost, but it still does not extrapolate to arbitrary corpora or models: the manual is synthetic and unambiguous, the staleness marker is computed by this system, and marker failure is untested. Costs include more local operations and latency. No claims are made of parameter self-evolution, full Bayesian RL, or general language-model improvement. Future work should connect this same process to real host AI and record frozen first answers, objective tool outcomes, feedback, reuse, and counterexamples round by round to test improvement on related tasks.

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## Appendix A. Implementation Mapping and Reproducible Experiments

The implementation mapping is as follows. Experience storage keeps the current version, parent version, and source event. Practice recording handles success, failure, and dispute transitions. Context composition and parsing retrieve candidate experience and enforce entity, version, and completeness admission. Backup and recovery preserve service state and evidence paths. The mathematical controls implement Beta updating, information gain, utility, and Thompson sampling. The paper explains the key modules, algorithms, constraints, and experimental definitions directly rather than replacing them with an undisclosed static ledger.

The reported experiments consist of deterministic mechanism replay, lifecycle controls, numerical controls, and a dream-effect comparison. Neither the main experiment nor the control experiments call a language model. The dream-effect comparison uses a host-supplied candidate template and produces evidence through real verification and recording interfaces; it is therefore not autonomous discovery or an independent model experiment. These results are mechanism evidence and cannot replace the unfinished confirmatory experiment with a real host AI.

## Appendix B. Additional Experimental Figures

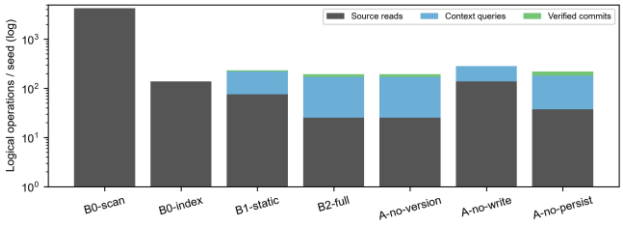



Figure 7. Evidence acquisition, retrieval, and writeback operations in the practice process.

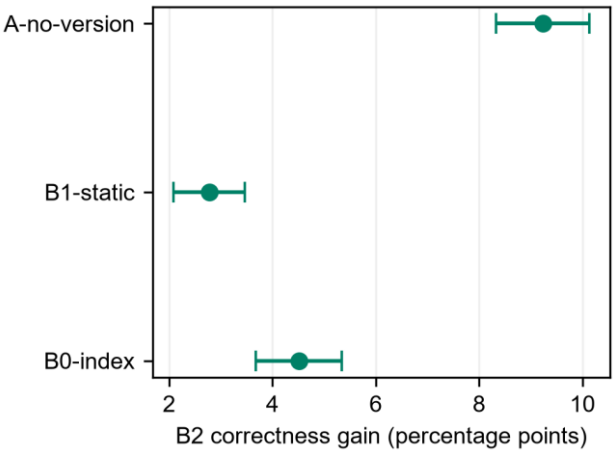



Figure 8. Seed-cluster paired effects.